THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 1 & 2: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Why should we never taste unknown chemicals? 1

(a) They are always sour **(b)** They have no taste
(c) They are costly **(d)** They may be harmful

2. Lime water is: 1

(a) basic **(b)** a strong acid **(c)** acidic **(d)** neutral

3. Heartburn can be relieved by taking: 1

(a) vinegar **(b)** a mild base such as an antacid
(c) lemon juice **(d)** tamarind water

4. Cabbage juice is used as an indicator because it: 1

(a) is neutral **(b)** changes colour with acids and bases
(c) has no colour **(d)** is sweet

5. A mixture of lemon juice and lime water in correct amounts becomes: 1

(a) red **(b)** more basic **(c)** neutral **(d)** more acidic

6. A person should not use an electric device near water because water: 1

(a) is an insulator **(b)** can conduct electricity
(c) is acidic only **(d)** is cold

7. The material of a bulb's filament is: 1

(a) aluminium **(b)** tungsten **(c)** iron **(d)** copper

8. Which element of the circuit controls current by gap? 1

(a) Switch **(b)** Filament **(c)** Terminal **(d)** Insulator

9. A good example of conduction in the human body is when: 1

(a) we eat food **(b)** current passes through wet skin
(c) we sleep **(d)** we read

10. Which part of a torch bulb connects to the cell terminal? 1

(a) the filament only **(b)** the glass bulb
(c) the reflector surface **(d)** the metal base and the tip

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 15–25 words each.

11. What happens to red rose extract when lemon juice is added? 2

12. Name two substances each that are acidic, basic and neutral. 2

13. What is meant by a closed circuit? 2

14. Why do we use copper in electric wires? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A field has become acidic. Name the substance a farmer should add and give the reason. 3

16. A battery is made of 3 cells of 1.5 V in series. What is its total voltage? 3

17. Which is more suitable to relieve a stomach ache: lemon juice or milk of magnesia? Why? 3

OR

State two uses of bases in daily life. 3

18. Why can pure water not be used to test for conduction in the same way as lemon juice? 3

OR

List the steps to light a bulb using a cell and two wires without a holder. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) How does knowledge of acids and bases help in cleaning? 5
(ii) Give two examples.

OR

(i) What is acid rain? 5
(ii) Effects on monuments and water bodies.
(iii) Two preventive steps.

20. (i) How do you separate conductors from insulators in a collection? 5
(ii) State two surprising conductors.

OR

(i) Name the symbols used for cell, battery, switch, bulb. 5
(ii) Draw a circuit with two bulbs in a line (describe).
(iii) Why are symbols helpful?

END OF QUESTION PAPER — Total: 40 Marks

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